Student Enrollment & Course Management System (SECMS)

Project Document

Salesforce Application

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1. Brainstorming & Ideation Phase

1.1 Introduction

SECMS is a Salesforce application that manages student enrollment and course information efficiently. It gives administrators, enrollment officers, faculty and students one centralized platform for student records, course offerings, enrollments, instructor assignments and reports.

1.2 Problem Statement

Managing student enrollment manually is time-consuming and error-prone. Searching, updating and generating reports from paper records or spreadsheets is difficult.

Existing Problems	Salesforce Solution
Manual student records	Student management
Scattered course data	Course management
Enrollment errors	Enrollment tracking
Duplicate data	Automated validation and flows
Difficult report generation	Reports and dashboards

1.3 Objectives

- Maintain student information electronically
- Provide an easy course enrollment process
- Maintain course and faculty details
- Reduce manual paperwork and data entry errors
- Provide quick searching and updating of records
- Maintain enrollment history
- Generate useful academic reports
- Provide secure access
- Improve the efficiency of academic administration

1.4 Proposed Idea

A centralized system that stores student, course, faculty and enrollment data, with easy registration, fast enrollment, quick search, better accuracy, reduced paperwork, secure data and easy reports.

2. Requirement Analysis Phase

2.1 Existing System

Data is kept in paper records, registers, spreadsheets and separate files. Disadvantages: time-consuming, more paperwork, duplication, difficult searching, more errors, difficult reports, security problems and difficult maintenance.

2.2 Functional Requirements (Scope)

Module	Functions
Student Management	Add, update, delete and view student details
Course Management	Add and manage courses
Faculty Management	Maintain faculty information
Enrollment Management	Enroll students in courses
Search	Search students and courses
Reports	Generate enrollment and student reports
Login	Secure administrator/user access

2.3 Hardware Requirements

Hardware	Minimum Requirement
Processor	Intel i3 or equivalent
RAM	4 GB
Storage	10 GB free space
Monitor / Keyboard / Mouse	Standard

2.4 Software Requirements

Software	Requirement
Operating System	Windows / Linux
Platform	Salesforce (Setup, Object Manager, Flows, Reports)
Browser	Chrome / Firefox / Edge
Alternative stack	Python / Java / PHP with HTML, CSS, JavaScript and MySQL

2.5 Security Requirements

- Username and password authentication
- Role-based access
- Database protection and input validation
- Restricted administrative functions
- Secure session management

3. Project Design Phase

3.1 System Architecture

User Login > Application Layer (Student, Course, Enrollment, Faculty and Report Management) > Database (Students, Courses, Faculty, Enrollments).

3.2 Entity Relationship

Entity	Attributes	Primary Key	Foreign Key
Student	Student_ID, Name, Email, Phone	Student_ID	-
Course	Course_ID, Course_Name, Credits	Course_ID	-
Enrollment	Enrollment_ID, Student_ID, Course_ID, Date	Enrollment_ID	Student_ID, Course_ID

Enrollment connects Student and Course, giving a many-to-many relationship.

3.3 Data Flow Diagram - Level 0

The whole system is one process, 0. SECMS. Students send registration and enrollment details and receive confirmation. Administrators send student and course details and receive reports.

3.4 Database Design

Student Table

Field	Type	Description
Student_ID	INT	Primary key
Student_Name	VARCHAR	Student name
DOB	DATE	Date of birth
Gender	VARCHAR	Gender
Email	VARCHAR	Email address
Phone	VARCHAR	Contact number
Department	VARCHAR	Department
Address	VARCHAR	Student address

Course Table

Field	Type	Description
Course_ID	INT	Primary key

Course_Name	VARCHAR	Course name
Department	VARCHAR	Department
Credits	INT	Course credits
Duration	VARCHAR	Course duration
Faculty_ID	INT	Assigned faculty

Faculty Table

Field	Type	Description
Faculty_ID	INT	Primary key
Faculty_Name	VARCHAR	Faculty name
Department	VARCHAR	Department
Email	VARCHAR	Email address
Phone	VARCHAR	Contact number

Enrollment Table

Field	Type	Description
Enrollment_ID	INT	Primary key
Student_ID	INT	Student reference
Course_ID	INT	Course reference
Enrollment_Date	DATE	Date of enrollment
Status	VARCHAR	Enrollment status

Admin Table

Field	Type	Description
Admin_ID	INT	Primary key
Username	VARCHAR	Login username
Password	VARCHAR	Login password (store encrypted)
Role	VARCHAR	User role

3.5 Input and Output Design

Student Registration Form: Student ID, Name, Date of Birth, Gender, Email, Phone, Department, Address (Submit, Reset). Course Enrollment Form: Student ID, Student Name, Course ID, Course Name, Enrollment Date, Status (Enroll, Cancel).

Student ID	Name	Department	Email	Course
101	Student A	B.Sc CS	studentA@email.com	Python
102	Student B	B.Sc CS	studentB@email.com	Java
103	Student C	BCA	studentC@email.com	Web Development

4. Project Planning Phase

Suggested plan based on the project modules. Adjust the durations to your college schedule.

Task	Description	Suggested Duration
Ideation	Problem statement, objectives, proposed idea	Week 1
Requirements	Scope, hardware, software, security	Week 1
Design	Architecture, ER, DFD, database design	Week 2
Data model setup	Create Student, Course, Faculty and Enrollment objects with fields	Week 3
Relationships and record types	Lookups, New Enrollment and Re-Enrollment record types	Week 3
Automation and security	Validation rules, flows, role-based access	Week 4
Reports and dashboards	Student, course, enrollment and faculty reports	Week 4
Testing	Test cases for all modules	Week 5
Documentation and demo	Final document, presentation, demonstration	Week 6

5. Project Development Phase

5.1 Student Management

Add, update, delete, view and search students. Created as the custom object Student in Salesforce Object Manager, with fields for ID, name, date of birth, gender, email, phone, department and address.

5.2 Course Management

Add, update, delete, view and search courses offered by the institution.

5.3 Faculty Management

Stores faculty ID, name, department, email, phone and course assigned. The Enrollment object has a lookup relationship to the Instructor object.

5.4 Enrollment Management

The main module. Record types: New Enrollment (first-time enrollments) and Re-Enrollment (student re-enrolls in a course). Process:

- Student login
- View available courses
- Select course
- Check course availability
- Confirm enrollment
- Store enrollment
- Display enrollment confirmation

5.5 Report Management

Reports: student list, course list, course-wise student list, student enrollment report, faculty report and enrollment status report.

6. Project Testing Phase

Test Case	Input	Expected Result	Status
Login	Valid username/password	Login successful	Pass
Login	Invalid password	Error message	Pass
Student Add	Valid details	Student added	Pass
Course Add	Valid details	Course added	Pass
Enrollment	Valid student/course	Enrollment successful	Pass
Search	Student ID	Student displayed	Pass
Report	Valid request	Report generated	Pass

7. Project Documentation Phase

7.1 Advantages

- Reduces paperwork and saves time
- Improves data accuracy and reduces duplication
- Centralized storage and quick searching
- Easier enrollment and report generation
- Better accessibility for authorized users

7.2 Limitations

- Requires a computer or mobile device
- Requires database maintenance
- Users need basic computer knowledge
- Internet connectivity needed for the online version
- Advanced security needs extra implementation

7.3 Future Enhancements

- Mobile application and online payment
- Email/SMS notifications
- Attendance, online examination and mark management
- Certificate generation and timetable management
- AI-based course recommendations and student performance analytics
- Cloud database

8. Project Demonstration Phase

8.1 Demo Flow

- Show the problem statement and solution overview
- Show the Student, Course, Faculty and Enrollment objects in Object Manager
- Register a new student
- Add a course and assign an instructor
- Enroll the student using the New Enrollment record type
- Show a Re-Enrollment example
- Open reports: course-wise student list and enrollment status
- Show login and role-based access

8.2 Conclusion

SECMS provides an efficient computerized solution for managing student and course activities. It reduces manual work and keeps student, faculty, course and enrollment records in one place. By automating registration and enrollment, it makes academic administration faster, more organized and easier to manage.